Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/07/2007
PI Name:	Yamauchi, Mitsuo Ph.D., D.D.S.		
Project Title:	Mechanisms of Bone Loss Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	27599	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mochida, Yoshiyuki Ph.D. (University of North Carolina)		
Grant/Contract No.:	NAG2-1596		
Performance Goal No.:			
Performance Goal Text:			
	The long-term goal of this research is to understand, on the molecular, cellular and biochemical levels, the mechanisms of bone loss caused by the condition of microgravity. Our underlying hypothesis is that cells in weight-bearing bones, when subjected to microgravity/skeletal unloading, synthesize and deposit post-translationally altered type I collagen that does not mineralize properly leading to bone loss. During the last grant period, by employing a rat tail suspension model, we found that skeletal unloading resulted in accumulation of collagen with elevated lysine hydroxylation and altered cross-linking pattern in weight-bearing bones and that the alterations were bone-region specific. In addition, the lysyl hydroxylase gene expression was significantly up-regulated in these bones. In this renewal, as a natural extension of these findings, we now propose two major objectives: 1. to further define the skeletal unloading-induced collagen alterations in weight-bearing bones by employing a rat-tail suspension model, and 2. to investigate whether or not		

Task Description:	elevated lysine hydroxylation of collagen is a cause of impaired collagen and poor mineralization in vitro. The first aim will be achieved by biochemical analysis, newly developed bioimaging and molecular technologies that are complementary to one another. The second aim will be achieved by evaluating the effects of overexpression of lysyl hydroxylase genes in osteoblastic cell line on a type I collagen molecule, cross-linking and mineralization in vitro. These extensive and complementary studies will hopefully provide insights into the mechanisms of bone loss occurred during spaceflight and would help develop its preventive strategy. To understand the mechanisms of bone loss/defects induced by skeletal unloading/microgravity. This study may help to develop strategies to prevent the bone loss that occurs during spaceflight/unloading.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objective of this study is to understand the mechanisms of bone loss/defects induced by skeletal unloading/microgravity. The study may help develop strategies to prevent the bone loss occurred during spaceflight/skeletal unloading.
Task Progress:	During the last grant year (no cost extension period), we mainly focused on collating data already generated. In an additional set of experiment, partially supported by this grant, we investigated the response of genes encoding collagen modifying enzymes (LH1-3, LOX, LOXL1-4) to BMP2-induced osteoblast differentiation using C2C12 cells. After BMP-2 treatment, osteoblast differentiation was confirmed by cell morphology and the expression pattern of bone markers. The expression of all isoforms of LH and LOX/LOXL except LH2a and LOXL2 were upregulated in a dose-dependent manner as the treatment time progressed. The expression of LH2a and LOXL2 were downregulated by BMP-2 with no dose dependency. The results indicate that the genes of LH and LOX isoforms differentially respond to BMP-2 induced osteoblast differentiation that may lead to the specific cross-linking pattern seen in bone collagen. Here again, the data suggest a critical role of specific collagen modifications/cross-linking in bone formation. During the course of characterization of collagen associated proteoglycans, partially supported by this grant, we have identified a gene encoding a novel member of small leucine-rich proteoglycans. We named one as "nephrocan" due to its high expression in kidney and characterized its interaction with transforming growth factor-beta.
Bibliography Type:	Description: (Last Updated: 08/11/2020)
Abstracts for Journals and Proceedings	Katafuchi M, Mochida Y, Atsawasuwan P, Sato H, Yamauchi M. "Podocan regulates TGF-ß activity. " Presented at the 84th Int Assc Dent Res, Brisbane, Austraia, June 28-July 1, 2006. J Dent Res 2006 Special Issue. , Jun-2006
Abstracts for Journals and Proceedings	Kaku M, Mochida Y, Atsawasuwan P, Parisuthiman D, Yamauchi M. "Gene expression of collagen modifying enzymes upon BMP-2 induced osteoblast differentiation." Presented at the 84th Int Assc Dent Res, Brisbane, Australia, June 28-July 1, 2006. J Dent Res 2006 Special Issue. , Jun-2006
Abstracts for Journals and Proceedings	Mochida Y, Parisuthiman D, Katafuchi M, Atsawasuwan P, Kaku M, Yamauchi M. "FLJ: A Novel Collagen-binding Small Leucine-rich Repeat Proteoglycan in Bone." Presented at the 28th Amer Soc Bone Miner Res, Philadelphia, Sep 15-19, 2006. J Bone Miner Res 2006 Sep;21:S81. , Sep-2006
Articles in Peer-reviewed Journals	Mochida Y, Parisuthiman D, Yamauchi M. "Biglycan is a positive modulator of BMP-2 induced osteoblast differentiation." Adv Exp Med Biol. 2006;585:101-13. PMID: 17120779 , Nov-2006
Articles in Peer-reviewed Journals	J Mochida Y, Parisuthiman D, Kaku M, Hanai J, Sukhatme VP, Yamauchi M. "Nephrocan, a novel member of the small leucine-rich repeat protein family, is an inhibitor of transforming growth factor-beta signaling." J Biol Chem. 2006 Nov 24;281(47):36044-51. PMID: 16990280 , Nov-2006
Articles in Peer-reviewed Journals	Yamauchi M, Shiiba M. "Lysine hydroxylation and cross-linking of collagen." Methods Mol Biol. 2008;446:95-108. http://dx.doi.org/10.1007/978-1-60327-084-7_7 ; PubMed PMID: 18373252 (Post-translational modifications of proteins: Tools for functional proteomics. Humana Press, Ed. C. Kannicht.) , Jan-2008